Sultan Qaboos University
College of Engineering
Office of Assistant Dean for
Industrial Training and Community Services

Weekly Report
Week #: 5
Date: 29/07/2026

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Supervisor	Full Name: Khalid AL-Habsi	Email:
	Training Company & Department: MODES & BULK AC	Tel./Mobile: 339642

Tasks Completed (Briefly Describe any tasks that you completed during this week):

• Learned Bulk AC Components.	•
• understanding maintenance procedure.	•
• Learned HVAC saftey cycle.	•

Tasks in Progress (if any):

• Learned chiller cycle.	•
•	•
•	•

Plan for next week (if different from this week):

• learn spilt AC system	•
• learn maintenance Procedure	•
•	•

Problems/Challenges/Recommendations (if any):

No major Problem we phase.

Supervisor Comments (your feedback is very important for our continuing improvement):

The student demonstrated good desPline and willings to learn.

Student:
I have read and understood the whole content of the Students' Training Manual.

Signature: ....

Supervisor:
I will do my best to guide the student towards achieving all required deliverables.

Signature: ....

Note: A copy of this report should be faxed to 24413416 or scanned and emailed to your SQU supervisor on a weekly basis. Originals should be placed in the Appendix of the **Final Report**.

WEEKLY TECHNICAL SUMMARY

HVAC Systems & Chiller Operations

Focus: AC Components & Chillers **Scope:** Maintenance & Safety **Date:** Current Training Week

This report details hands-on experience, maintenance procedures, HVAC safety protocols, and operational understanding of bulk AC components and commercial chiller cycles acquired during this week's technical sessions.

1. BULK AC COMPONENTS & MECHANICS

Understanding component interaction is essential for systematic troubleshooting and performance evaluation.

Component	Role	Functional Description
Compressor	PRESSURE/FLOW	Acts as the heart of the refrigeration circuit. Compresses low-pressure, low-temperature vapor into a high-pressure, high-temperature gas.
Condenser Coil	HEAT REJECTION	Rejects absorbed indoor heat to the ambient outside air. As heat leaves, high-pressure gas condenses into high-pressure liquid.
Metering Device (TXV)	PRESSURE DROP	Regulates refrigerant flow into the evaporator. Creates a sudden pressure drop that cools the refrigerant significantly.
Evaporator Coil	HEAT ABSORPTION	Absorbs thermal energy from indoor air into the cold liquid refrigerant, vaporizing it while cooling and dehumidifying space air.

2. PREVENTIVE MAINTENANCE PROCEDURES

Routine preventive maintenance protects system efficiency, prevents unexpected mechanical failures, and extends equipment operational life:

- **Visual & Structural Inspection:** Checked fan blade balances, belt tension, casing integrity, and electrical wire insulation for heat degradation or vibration wear.
- **Coil Cleaning Protocols:** Applied appropriate foaming cleaner solutions to condenser coils, washing in the reverse direction of airflow to clear debris and restore optimal heat transfer.
- **Air Filtration Management:** Evaluated static pressure drops caused by clogged media and established routine replacement schedules for pre-filters and final filters.
- **Electrical & Refrigerant Diagnostics:** Checked operating compressor amperage against RLA ratings and verified refrigerant subcooling/superheat values.

3. HVAC SAFETY CYCLE & WORKING PROTOCOLS

CRITICAL SAFETY PROTOCOLS LEARNED

- **Lockout / Tagout (LOTO):** Always isolate electrical feed breakers and test using a calibrated multimeter to confirm zero energy state before servicing control boards or compressors.

- **PPE Standard Compliance:** Non-negotiable usage of voltage-rated insulated gloves, safety glasses with side shields, and steel-toe boots during all troubleshooting tasks.
- **Refrigerant Handling:** Strict adherence to EPA recovery standards—preventing direct skin contact burns from liquid refrigerant and avoiding asphyxiation hazards in tight mechanical rooms.

4. COMMERCIAL CHILLER CYCLE BREAKDOWN

Unlike standard direct-expansion (DX) systems, commercial chillers utilize water as a primary heat transfer fluid to cool large facilities.

Stage 1	Stage 2	Stage 3	Stage 4
Evaporator (Barrel)	**Compression**	**Condenser**	**Expansion**
Warm building water transfers heat into cold refrigerant, creating **Chilled Water** (~44°F) sent back to building AHUs.	Vaporized refrigerant enters the centrifugal/screw compressor, elevating pressure and temperature for heat rejection.	Heat transfers into a secondary **Condenser Water** loop, which pumps thermal load to outdoor cooling towers.	Refrigerant passes through expansion valve, dropping in pressure and temperature before re-entering the evaporator barrel.

Key Takeaways & Reflections

HVAC systems revolve around fundamental principles of thermodynamic heat transfer and pressure differentials. Mastering the relationship between pressure, temperature, and fluid flow makes diagnosing complex chiller and DX failures straightforward. Continued focus will center on advanced multimeter diagnostics and manifold gauge reading analysis on active chiller barrels.